

Tactile Communication on the Palm: A Study of Sensory Perception and Signal Design

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Abstract

Palm-based tactile communication has the potential to enhance non-verbal interaction, assistive technologies, and immersive experiences. This study investigates how well-designed vibration signals on specialized surfaces influence human perception, intending to establish a structured approach for palm-based tactile communication. We conducted user experiments to evaluate participants' ability to perceive and differentiate tactile stimuli across various surface designs and configurations, including simultaneous and serial modes. The system comprises electronic hardware and an interactive dashboard that controls the vibration device while guiding participants through structured interactions. The results indicate that gaps between actuators and palm areas strongly impact human sensation. Our findings provide critical insights for designing effective tactile communication systems using palm-based tactile signals for effective information transmission. It also paves the way for integrating vibration signals with AI, enhancing non-verbal communication and supporting diverse groups, including individuals with disabilities.

Keywords

Tactile Communication, Vibration Signals, Palm Sensitivity, Assistive Device

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1 Introduction and Background

The human palm is one of the most sensitive areas of the body, capable of detecting subtle variations in pressure, temperature, and texture [5, 10]. This high sensitivity makes it an ideal medium for *speechless interaction*, offering a novel communication channel that complements speech, text, and images. Palm-based tactile communication is gaining traction, particularly in assistive technologies, which enhance accessibility and user interaction.

A prominent example is the *LORM alphabet*, a tactile language designed for individuals with deafblindness, where precise touches and swipes on the palm represent different letters [14, 21]. The success of such systems underscores the potential of *palm-based communication* in bridging sensory gaps and facilitating richer non-verbal interactions.

1.1 Vibration Signals for Tactile Communication

Vibration signals have been widely used to transmit information across various applications. The Emma Watch [22] uses vibrations to help Parkinson's patients control tremors, while Oroszi et al. (2020) [20] demonstrated how vibrations effectively stimulate tactile receptors. Researchers have recently explored vibrating wearables for navigation, gesture guidance, and silent notifications [4, 11, 23]. However, many existing systems lack adaptability, transmitting only basic information and limiting their use in complex tactile communication. Due to its high tactile sensitivity, the palm offers an ideal platform for developing multi-dimensional tactile signals. Recent studies have focused on optimizing vibration-based interactions for discreet smartphone alerts [4], secure notifications [11],

and assistive tools for the visually impaired [6]. However, most systems still rely on simple stimuli, preventing detailed information encoding.

Tactile communication extends beyond Braille-based interfaces, as demonstrated by previous studies investigating the effectiveness of wearable vibration-based assistive devices. Such as vibrotactile belts and wristbands, in aiding spatial awareness and mobility for blind users [17, 26]. Meanwhile, mid-air haptic technology enabling non-contact feedback, like ultrasonic haptics, allows users to sense spatial vibrations, enhancing assistive communication [9, 24]. Integrating these technologies could enable future systems to simulate complex tactile patterns without direct contact.

1.2 Palm-Based Tactile Feedback Systems

Several studies have explored advanced tactile systems using vibrotactile matrices. Jeunet et al. (2015) [12] and Borja et al. (2018) [2] developed a haptic glove with palm-based vibrotactile motors for virtual interaction. Another study in Seoul [25] optimized user interface design based on palm sensitivity, though it overlooked the effects of pressure distribution on perception accuracy. A horizontal vibration surface provides better signal reception than curved or complex structures [28], ensuring even force distribution and comfort. Elsayed's research [7] found optimal palm sensitivity at 100–300 Hz, peaking around 200 Hz, but did not explore how to distinguish vibrations across multiple palm locations, limiting its applicability in multi-signal tactile interfaces.

1.3 Study Objectives

This research investigates palm-based tactile communication by evaluating multi-point vibration signals to enhance non-verbal interaction. We designed a system to deliver precise vibration patterns across the palm, enabling participants to perceive and interpret the signals. We then conducted a study to assess participants' ability to recognize and differentiate these patterns under varying conditions. The findings aim to improve tactile interaction designs by enhancing the detectability and differentiation of vibration signals, optimizing patterns for effective tactile encoding, and exploring the potential integration of palm-based haptic feedback into assistive technologies, virtual reality, and advanced communication systems for accessible and inclusive interaction.

2 Methods

2.1 Surface Design

The device surface consists of a foam rubber base for cushioning and a durable polyurethane leather top. Integrated LRA coin motors provide vibration feedback. The flat surface ensures even contact and uniform vibration transmission to the skin [8], enhancing tactile clarity and minimizing signal distortion. Using rubber foam reduces unwanted resonances, as hard materials can amplify certain frequencies and distort the vibration response [16]. The surfaces are designed with different dimensions, focusing

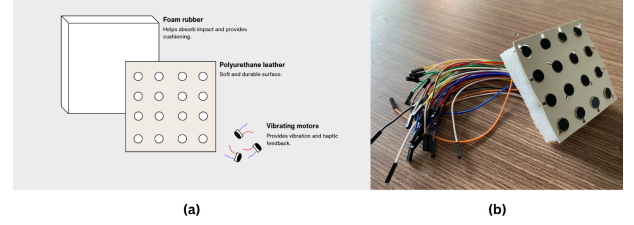


Figure 1: (a) Visualized design of vibrating surface (b) Real version of vibrating surface

on investigating the signal transmission across regions like edges, center, and other areas (see Figure 1).

According to the study by Seyyed J. M. et al., the adult men's palm breadth (measured from the outer edge of the 2nd metacarpal joint to the 5th metacarpal joint) ranges from 7.84 to 8.73 cm [18]. However, palm size varies significantly depending on race and nationality. Previous work has shown that the average palm breadth of adult men is about 7.92 cm [18]. Additionally, the study by the Vietnam Military Medical Academy also showed that the length of the hand (measured from the palmar crease to the corresponding fingertip) is approximately 9.80 cm [1]. Based on this information, we designed the surface to ensure that the surface is right in the center of the participant's palm, with the following distances between the vibration motors; dimensional parameters will be indicated in Table 1.

2.2 Box Design

To support the fixation of the vibration surface, we have designed special boxes to hold the surface firmly during the experiment (see Figure 2). The box surface is designed to have a long space, creating comfortable support for the participants and reducing hand fatigue during long procedures. The top of the box has a square, where the surface containing the vibration motors is placed. The small grooves on the surface are placed to thread the wires connecting the vibration motors without affecting the flatness and signal transmission efficiency. Specifically, the bottom of the box is designed to be flexible, allowing experimenters to slide it in and out easily, enabling quick replacement of the ESP32 board when it is necessary.



Figure 2: Prototype device with 3D-printed

Table 1: Optimized Gap Design (in cm)

Box name	Width	Length	Gap
Box 1	7.00	7.00	1.00
Box 2	6.40	6.40	0.80
Box 3	5.80	5.80	0.60
Box 4	5.20	5.20	0.40

2.3 Dashboard design

The system integrates with a software dashboard, providing a platform that allows the experiment host to configure the hardware for various vibration signals. The hardware comprises an embedded system that receives signals from the client side while incorporating optimized design models to ensure the accuracy and integrity of the experimental results. The dashboard prepares vibration patterns for each experiment and serves as a screen that supports participants

in interacting with the device, such as starting the experiment, moving to the next experiment, or displaying a countdown. The ESP32 is configured with a baud rate of 115200 and transmits an object as the configuration for each trial. As a result, the duration of the hardware activation from the dashboard ranges between 10ms and 14ms. The process of participants interacting with the software is presented in the following section.

3 User Study

3.1 Experiment Procedure

3.1.1 Experimental Setup. The process involved 104 trials, during which participants were asked to experiment for 30 to 35 minutes. Before starting, all participants were required to read the description of the experimental procedure. If they agreed, they were asked to sign a consent form. Participants sat on a chair with a suitable height, facing a laptop that displayed instructions for each pattern. Each box was placed to the left and replaced sequentially for each section. To start the trial, participants pressed the "Spacebar" on the keyboard, with a random delay of 1 to 3 seconds to prevent bias. After each vibration, participants wrote their answers on the multiple choice questions sheet on an iPad and then pressed "C" to move on to the next trial, repeating until the end of the experiment.

Participants had 2-minute breaks while experimenters prepared the next setup. After completing all sections, participants joined a semi-structured interview, discussing their perceptions, confidence in identifying simultaneous vibrations, and challenges faced. They were asked to share which box provided the clearest sensations and their observations regarding the effectiveness of the stimuli. Figure 3 shows an overview of the experimental procedure.

3.1.2 Participants. Ten participants were selected based on the absence of hand deformities or calluses resulting from labor or congenital conditions. All participants were students and office workers, as they are less likely to have hand-related conditions that might affect tactile sensitivity. Before participating, all individuals were required to provide informed consent. Figure 4 presents images of three out of the ten selected participants.



Figure 3: Experimental Procedure Timeline

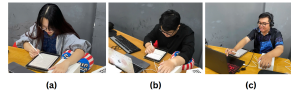


Figure 4: Participants testing different vibration patterns

3.2 Experiment Design

To comprehensively evaluate participants' sensations, we carefully designed four cases in the experiments. The experiment involves different vibration patterns across 4 boxes with motor gaps. Each box contains 4 cases. Participants performed a counterbalanced order of boxes, cases, and vibration patterns. These cases aim to assess palm perception and serve as a foundation for real-world applications. Understanding user perception enables the design

of more effective vibration signals for information transmission. Figure 5 shows the visualization of these 4 cases. Specifically, we describe each case in detail and explain the corresponding evaluation criteria.

The evaluation consists of four distinct cases designed to analyze various aspects of vibration-based tactile feedback. In **Case 1**, a single vibration motor is activated for each trial to eliminate noise signals and provide the most recognizable feedback. Each vibration lasts 2 seconds at an intensity of 235 Hz [27], with 16 trials performed in random positions. **Case 2** involves simultaneously activating two adjacent motors to assess the distinction between overlapping signals. This test maintains the same duration and intensity as **Case 1** for consistency. **Case 3** introduces a gap by activating two motors with a non-vibrating motor in between, allowing the evaluation of how spatial distance influences signal perception and whether confusion arises from vibration spillover or positional bias. Finally, **Case 4** simulates horizontal and vertical swipe motions on the palm by activating each motor with a 100ms delay. Adjusting the activation delay to 100ms enhances the perceived swipe feel, helping participants to clearly distinguish the sensations [13]. Regarding the hardware parameter of our vibrating actuators (LRA VG0832022D), the motor's rise time is 50ms, so we configured the parameter to 50ms, to ensure the total delay will be 100ms. Furthermore, we aim to create dynamic motion, so it is necessary to set the duration of each motor's vibration to be short. Our assumption for the duration is 100ms, so its performance will be evaluated through the experimental results.

4 Experimental Results

The comparison between participants' responses and the correct answers revealed a low accuracy of 25.00%. However, a deeper analysis of the participants' incorrect answers uncovered patterns of common perceptions, which accounted for 57.50% of perceptions. The participants' perceptions were grouped into four common perceptions, including adjacent motor misunderstanding, direction misunderstanding, adjacent middle motor misunderstanding, and missing motor perception. The heatmaps in Figure 6 showed the distribution and frequency of these perceptions, highlighting the most common perceptions made by participants during the task. Each subfigure's left side shows the given sample signal, while the right side displays the participant's perceived and reported signal.

The observed perceptions in motor sensitivity experiments were summarized as follows. Participants often experienced **Adjacent motor Misunderstanding**, where they mistakenly associated vibrations from nearby motors, either horizontally or vertically adjacent, with the correct motor. Additionally, **Direction Misunderstanding** occurred when participants confused horizontal and vertical directions, leading to misidentification of the vibration's orientation. Another common issue was **Adjacent Middle motor Misunderstanding**, where participants failed to perceive the gap between two non-adjacent vibrating motors, instead assuming that

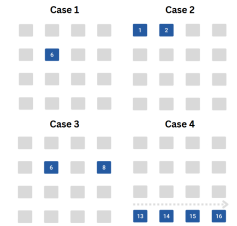


Figure 5: Visualization of four vibration cases

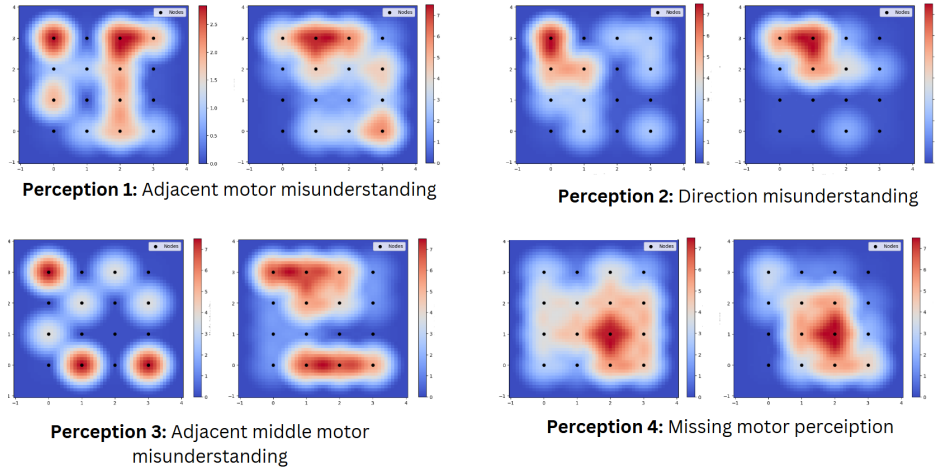


Figure 6: Four Common Participants' Perceptions

Table 2: Comparison of metrics across BOX 1 to BOX 4.

	BOX 1	BOX 2	BOX 3	BOX 4
Accuracy (%)	39.97	14.08	21.36	19.54
Accuracy + Perception 1 (%)	63.42	29.54	37.27	36.36
Confidence (avg.)	3.77	3.68	3.63	3.78
Confidence (std)	0.91	1.01	0.95	0.99

all three motors (including the gap) were active. Lastly, **Missing motor Perception** happened when participants failed to sense a vibration pattern fully, missing the contributions of one or two motors in a sequence. These perceptions highlighted challenges in accurately interpreting motor vibration patterns. Additionally, based on the correct answers, combined with common perceptions and confidence levels, we may answer the research questions to establish a theoretical basis for palm vibration perception.

4.1 Research Finding 1: Minimum Gap Threshold for Perceiving Differences among Vibration Motors

To determine the optimal distance between two motors for human perception, we examine the precision and confidence of each box, which vary with the gaps between vibrating actuators. Table 2 presents the precision and confidence values for the four configurations. Box 1 (1.00 cm gap between actuators) achieves the highest accuracy (39.97%) and perception-1-inclusive accuracy (63.42%), indicating that a 1.00 cm motor gap is optimal for the average palm size of Vietnamese adults. Perception 1, caused by neighboring motor misunderstandings, does not significantly impact overall performance. Box 1 also shows the highest confidence average (3.77) with the lowest standard deviation (0.91). The results suggest that greater gaps might enhance vibration perception. Yet, Pearson correlation analysis ($p > 0.05$) shows no significant link between distance and accuracy. This indicates that simply increasing motor spacing does not guarantee better perception. Instead, optimizing vibration

placement based on hand size, receptor density, and structure is crucial for accuracy. This highlights that the key is an optimized gap, not just the minimum gap. Another surprising finding, Box 4 (0.40 cm gap), despite having lower accuracy (19.54%), exhibits the highest confidence average (3.78), highlighting a short correlation between accuracy and confidence. However, Pearson correlation tests found no significant correlation between them ($p > 0.05$).

4.2 Research Finding 2: Most/Least Sensitive Palm Areas

During this study, we aim to identify the areas of the palm that are challenging and simple for participants to recognize. These results help us better understand how participants respond to vibration signals in different areas. We assume that areas such as the palmar concavity or the left edge of the palm are more difficult to perceive because they are not flat or have fewer sensory receptors. However, the results of the data analysis do not necessarily support this assumption.

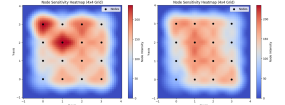


Figure 7: Left: Ground truth motor sensitivity. Right: Participant responses show discrepancies in perception within all trials.

Based on the heatmaps in Figure 7, the left shows the correct answers, while the right represents participants' responses. The central region is more prone to confusion, likely due to sensory overlap caused by the proximity of vibration points. In contrast, the peripheral regions, particularly the left and right edges of the palm, exhibit higher accuracy, reflecting better perception in these areas.

Additionally, after the experiential procedure, we conduct a semi-structured interview. All participants agree that the second and third rows (middle of the palm) are the most confusing, making

them feel less confident when choosing the perceived vibrating areas. Specifically, over **40.00%** of participants identify motors 6 and 7 (the middle two actuators of row 2) as the most confusing.

Table 3 visualizes the results for 4-motor and 2-motor patterns. The accuracy is calculated as the percentage based on the user's total responses for a trial. It states that the first row (1-2-3-4) and the last row (13-14-15-16) achieve the highest (**42.50%**), making them the easiest to perceive. Confidence scores are consistently high (**mean > 3.50**), with the highest scores for Horizontal top (**4.00**) and 1-2 (**4.18**). A low standard deviation (**0.58–0.89**) further indicates stability and consistency in participants' responses.

4.3 Research Finding 3: Ability to Distinguish Vertical and Horizontal Vibrations

We analyze data from case 2 and case 4 (simultaneous adjacent double vibration and serial swipe vibration, respectively) to determine which direction—horizontal or vertical—is better recognized in vibration patterns. Accuracy, defined as the percentage of correct responses, is higher for horizontal orientations (**31.87%**) compared to vertical orientations (**20.00%**). Overall accuracy is low, as many participants experience perception 4, where signals across the vibrating band are partially missed. However, incorporating perception 4 into correct responses raises the accuracy for horizontal signals to nearly **70.00%** and for vertical signals to nearly **50.00%**. This suggests that while signals are not perfectly perceived, responses often align with expected patterns. Additionally, participants show higher confidence in horizontal directions, with a mean confidence score of **3.88** compared to **3.56** for vertical directions.

In addition, Perception 3—confusion between vertical and horizontal directions—is significantly higher for the vertical direction (**75.00%**) than for the horizontal direction (**21.87%**). This suggests that participants are more likely to misinterpret the vertical direction as horizontal, reinforcing that the horizontal direction is easier to perceive and less confusing than the vertical direction. To confirm the effect of vertical and horizontal perception, we conducted a paired t-test, which is appropriate as the dependent variable (perception accuracy) is measured on a continuous scale, and the same group of participants ($n = 10$) was tested under both conditions, ensuring paired data. All assumptions of the t-test were checked and made sure that they were satisfied before the test was run [15]. The results show a significant difference in mean scores between the two conditions $t(9) = 9.78$, $p < 0.0005$, indicating that participants perceived horizontal vibrations much more accurately than vertical ones. Specifically, the mean score for the horizontal direction is 10.20 ± 1.14 , while the mean score for the vertical direction is 6.40 ± 2.07 . The large effect size (Cohen's $d = 2.27$) further supports this difference. Additionally, the higher standard deviation in the vertical direction suggests greater variability in participants'

Table 4: Comparison of Horizontal and Vertical Metrics

	Horizontal	Vertical
Swiping perception (%)	56.90%	43.08%
Accuracy (%)	28.12%	18.75%
Perception 4 (%)	44.70%	
Confidence (avg.)	3.73	3.44

responses, reflecting inconsistencies in their perception of vertical signals. These findings confirm that the horizontal direction is significantly easier to perceive and interpret. Besides, the higher standard deviation for vertical signals indicates greater variability due to participants' inconsistent experiences. These findings clearly demonstrate that the horizontal direction yields better results.

Our study also evaluates the potential of swiping signals generated by sequentially activating vibration signals with short delays and durations, specifically in Case 4, to identify the most effective directions and positions for participant accuracy, as detailed in Table 4. In the swiping motion tests, **79.06%** of trials are perceived as dynamic, specifically as swiping motions, with horizontal swipes being more frequently recognized (**56.90%**) compared to vertical swipes (**43.08%**). This demonstrates that the chosen parameters of a 100ms delay and a 100ms duration effectively create a clear and perceivable swiping motion for users. Horizontal swipes also demonstrate higher accuracy (**26.25%**) than vertical swipes (**18.75%**), and participants report greater confidence for horizontal swipes (**3.73**) versus vertical swipes (**3.44**). These results confirm that the horizontal orientation is easier for participants to perceive and respond to. Additionally, rows 1 and 4, and columns 1 and 4 on the palm yield the best responses, aligning with the hypothesis proposed in Research Finding 2.

However, when considering the accuracy according to the statistical table, the results are still not high. Further analysis reveals a common perception, referred to as Perception 4. According to the statistics, nearly **45.00%** of the common perceptions in Case 4 are related to this phenomenon. This highlights the need to address factors such as frequency, duration, and motor spacing, which may contribute to this misunderstanding in future studies.

4.4 Research Finding 4: Participants Perception of a Gap Between Two Simultaneously Vibrating Motors with One Middle Motor Left Idle

In Case 3, where trials are designed with two motors separated by one inactive motor vibrating simultaneously, we find a common perception: participants tend to confuse two vibration signals as one horizontal or vertical band, or they do not perceive that there is an unactivated motor between the two active ones. This finding is particularly noteworthy, as illustrated in Figure 8.

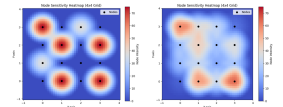


Figure 8: Left: Ground truth motor sensitivity. Right: Participant responses show discrepancies in perception of case 3.

Specifically, according to the statistics of **Perception 3**, approximately **64.30%** of trials show that participants fail to perceive the gap between the two activated motors, instead reporting the sensation of two adjacent motors vibrating. Moreover, in **35.70%** of trials, participants believe that the entire row or column containing the two activated motors are vibrating. We observe that the smaller the gap between the boxes, the more likely this occurs. For instance, Box 1 (1 cm) has the highest percentage of responses (**31.30%**), followed by Box 2 (0.60

Table 3: Performance metrics for different areas.

Direction	Motors	Accuracy (%)	Confidence (avg.)	Confidence (STD)
Horizontal top	1-2-3-4	42.50	4.00	0.87
Vertical right	4-8-12-16	32.50	3.58	0.89
Horizontal bot	13-14-15-16	42.50	3.92	0.82
Vertical left	1-5-9-13	20.00	3.50	0.87
Horizontal top	1-2	45.00	4.18	0.77
Vertical left	1-5	35.00	3.93	0.58
Horizontal top	3-4	42.50	3.79	0.72

cm) at **25.30%**, Box 3 (0.80 cm) at **23.20%**, and Box 4 (0.40 cm) at **20.20%**. This indicates a clear trend of decreasing accuracy as the box size and spacing decrease.

Participants demonstrate a high confidence level, averaging **3.88** with a standard deviation of **0.89**, despite significant confusion in perceiving the vibration signals. This indicates that participants are often unaware of their misperceptions, believing their answers to be correct.

5 Conclusion, Discussion, and Future Work

Our experiments, conducted with 10 participants with average Vietnamese hand sizes, indicated a low accuracy rate of 25.00%, attributed to shared misunderstandings regarding vibration position, direction, and incomplete signal perception. The palm's concave regions (positions 6, 7, 10, and 11) caused the most confusion and uncertain confidence. In addition, box 1 achieved the highest accuracy and participants' confidence. In contrast, Box 4, while showing low accuracy, surprisingly achieves high confidence, emphasizing a loose correlation between accuracy and confidence. Besides, participants demonstrated a better perception of horizontal vibrations than vertical ones. Moreover, the results of swiping trials support the effectiveness of the 100ms delay and 100ms duration parameters in enhancing the clarity of dynamic motion, particularly in dynamic tests. Additionally, trials involving two vibrating motors separated by a non-vibrating middle motor often led to misunderstanding, with participants perceiving them as either a row or column of vibrations. It indicates that overcoming and utilizing these common perceptions is necessary to optimize modern vibration systems.

Despite these findings, the current study has limitations in exploring vibrational rhythmic patterns. The weak correlation between accuracy and confidence suggests the need for a deeper analysis of cognitive factors affecting user perception and recognition. Moreover, participants struggled to differentiate signals in concave palm areas, leading to misinterpretation. These factors highlight the need for further optimizations in both hardware and signal processing.

In addition, we can take advantage of the concave area of the hand. It can utilize its confusion to increase the sensitivity of the tactile warning system, turning the multidirectional vibration here into a special signal for emergencies or requiring high attention. Besides, with participants' difficulty in noticing inactive middle motors (case 3), it can improve the device by activating only the first and last motors in a row or column, instead of all, to achieve the same effect while saving energy. Additionally, for the practical applications, we plan to miniaturize components, use lightweight

materials, and consolidate wiring with flex-circuit or ribbon cable while implementing wireless protocols like HTTP or MQTT for signal reception.

To help participants distinguish the vibration signal in the concave area from other areas on the palm, we propose increasing the intensity of the surrounding motors to create a clear contrast. Similarly, we expect to improve the remains of common perceptions, such as missing motor or direction confusion. We aim to conduct additional studies to optimize duration and intensity, enhancing human perception and overall tactile experience. Furthermore, in the future, we plan to conduct a deeper analysis of the factors influencing participants' confidence to understand how these factors contribute to the participants' perception and decision-making, especially since the results show a short correlation between accuracy and confidence. Additionally, we aim to assess the Sense of Agency (SoA) with a similar methodology as in [3], to explore its influence on perception and decision-making. Finally, while this study focuses on palm-based tactile communication and its applications, several promising research directions can further enhance this field. One avenue worth exploring is the integration of multisensory interactions to enrich user experiences. Harnessing findings from crossmodal correspondence research, tactile stimuli can be combined with other sensory modalities, such as auditory, visual, olfactory, and gustatory cues, to improve perception, information encoding, and user engagement [19].

In conclusion, the theoretical foundation of palm sensation enables novel communication channels, particularly benefiting individuals with disabilities. Integrating AI, we propose a speech-to-vibrator system to convert voice into vibration signals and emotion classification to detect emotions from tone, facilitating communication for the deafblind. Additionally, in VR/AR, gesture recognition, eye-tracking AI, and haptic feedback can enhance immersive interactions.

References

- [1] Tran Anh, Pham Van, and Vu Tung. 2022. Assessment of the relationship between hand size and upper limb size, standing height, and weight of students at Thai Binh University of Medicine and Pharmacy. *Vietnam Military Medical University* 47 (September 2022), 44–58. <https://doi.org/10.56535/jmpm.v47i6.50>
- [2] E. F. Borja, D. A. Lara, and W. X. Quevedo. 2018. Haptic Stimulation Glove for Fine Motor Rehabilitation in Virtual Reality Environments. In *Lecture Notes in Computer Science*. Springer. https://link.springer.com/chapter/10.1007/978-3-319-95282-6_16 Available online: SpringerLink.
- [3] Patricia Ivette Cornelio Martinez, Silvana De Pirro, Chi Thanh Vi, and Sriram Subramanian. 2017. Agency in Mid-air Interfaces. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 2426–2439. <https://doi.org/10.1145/3025453.3025457>

- [4] Kaiyan Cui, Qiang Yang, Yuanqing Zheng, and Jinsong Han. 2023. mmRipple: Communicating with mmWave Radars through Smartphone Vibration. 149–162. <https://doi.org/10.1145/3583120.3586956>
- [5] J. Dargahi and S. Najarian. 2004. Human tactile perception as a standard for artificial tactile sensing—a review. *The International Journal of Medical Robotics and Computer Assisted Surgery* 1, 1 (2004), 23–35. <https://doi.org/10.1002/rcs.3>
- [6] Swagat Das, Sonal Patro, Ritika Das, and Antara Mishra. 2019. Arduino based Safety Device for the Visually Challenged. In *2019 International Conference on Communication and Signal Processing (ICCSP)*. 0601–0605. <https://doi.org/10.1109/ICCSP.2019.8697987>
- [7] H. Elsayed, M. Weigel, F. Müller, and G. Ibrahim. 2023. Understanding stationary and moving direct skin vibrotactile stimulation on the palm. *arXiv preprint arXiv:2302.08820* (2023). <https://arxiv.org/pdf/2302.08820> Available online: arXiv.
- [8] Amon Gasagara, Wuyin Jin, and Angelique Uwimbabazi. 2020. Modeling of Vibration Condition in Flat Surface Grinding Process. *Shock and Vibration* 2020, 1 (2020), 3069895. <https://doi.org/10.1155/2020/3069895>
- [9] Orestis Georgiou, William Frier, and Oliver Schneider. 2022. *User Experience and Mid-Air Haptics: Applications, Methods, and Challenges*. 21–69. https://doi.org/10.1007/978-3-031-04043-6_2
- [10] N. H. H. Mohamad Hanif. 2015. *Sensory feedback for artificial hands*. Ph.D. Dissertation. University of Southampton. <https://eprints.soton.ac.uk/391380/> Available online: Eprints Soton.
- [11] Long Huang and Chen Wang. 2021. Notification privacy protection via unobtrusive gripping hand verification using media sounds. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (MobiCom '21)*. ACM, 491–504. <https://doi.org/10.1145/3447993.3483277>
- [12] Camille Jeunet, Chi Vi, Daniel Spelmezan, Bernard N’Kaoua, Fabien Lotte, and Sri-ram Subramanian. 2015. Continuous Tactile Feedback for Motor-Imagery Based Brain-Computer Interaction in a Multitasking Context. In *Human-Computer Interaction – INTERACT 2015*, Julio Abascal, Simone Barbosa, Mirko Fetter, Tom Gross, Philippe Palanque, and Marco Winckler (Eds.). Springer International Publishing, Cham, 488–505.
- [13] Ricardo Jota, Albert Ng, Paul Dietz, and Daniel Wigdor. 2013. How fast is fast enough? a study of the effects of latency in direct-touch pointing tasks. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Paris, France) (CHI '13)*. Association for Computing Machinery, New York, NY, USA, 2291–2300. <https://doi.org/10.1145/2470654.2481317>
- [14] E. Kasimova and R. Kutlubaeva. 2019. Teaching The Lorm Alphabet To Adults With Deaf-Blindness Acquired With Age. In *European Proceedings of Social and Behavioural Sciences*. European Proceedings. <https://doi.org/10.15405/epsbs.2020.11.53> Available online: European Proceedings.
- [15] Laerd Statistics. n.d.. Independent t-test using SPSS Statistics. <https://statistics.laerd.com/spss-tutorials/independent-t-test-using-spss-statistics.php>. Accessed: 2025-02-23.
- [16] Tommaso Lisini, Sara Marullo, Nicole D’Aurizio, and Domenico Prattichizzo. 2022. Discriminating different materials by means of vibrations. *Frontiers in Manufacturing Technology* 2 (08 2022), 939755. <https://doi.org/10.3389/fmtec.2022.939755>
- [17] Zhuoluo Ma, Yue Liu, Dejiang Ye, and Lu Zhao. 2019. Vibrotactile Wristband for Warning and Guiding in Automated Vehicles. (2019), 1–6. <https://doi.org/10.1145/3290607.3312819>
- [18] Seyyed Mirmohammadi, Amir Mehrparvar, Mehrdad Mostaghaci, Mohammad Hossein Davari, Maryam Bahaloo, and Samaneh Mashtizadeh. 2015. Anthropometric Hand Dimensions in a Population of Iranian Male Workers in 2012. *International journal of occupational safety and ergonomics : JOSE* 22 (12 2015), 1–17. <https://doi.org/10.1080/10803548.2015.1112108>
- [19] Marianna Obrist, Carlos Velasco, Chi Thanh Vi, Nimesha Ranasinghe, Ali Israr, Adrian D. Cheok, Charles Spence, and Ponnampalam Gopalakrishnakone. 2016. Touch, Taste, & Smell User Interfaces: The Future of Multisensory HCI. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (San Jose, California, USA) (CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 3285–3292. <https://doi.org/10.1145/2851581.2856462>
- [20] T. Oroszi, M.J.G. van Heuvelen, C. Nyakas, and E.A. van der Zee. 2020. Vibration detection: its function and recent advances in medical applications. *F1000Research* 9 (17 Jun 2020), F1000 Faculty Rev–619. <https://doi.org/10.12688/f1000research.22649.1>
- [21] O. Ozioko, W. T. Navaraj, N. Yogeswaran, and R. Dahiya. 2018. Tactile Communication System for the Interaction Between Deafblind and Robots. *IEEE Transactions on Robotics* (2018). <https://eprints.gla.ac.uk/201386/1/201386.pdf> Available online: IEEE Xplore.
- [22] A. Pacheco, T.A. van Schaik, N. Paleyes, M. Blacutt, J. Vega, A.R. Schreier, H. Zhang, C. Macpherson, R. Desai, G. Jancke, and L. Quinn. 2023. A Wearable Vibratory Device (The Emma Watch) to Address Action Tremor in Parkinson Disease: Pilot Feasibility Study. *JMIR Biomedical Engineering* 8 (23 Oct 2023), e40433. <https://doi.org/10.2196/40433>
- [23] Martin Pielot, Niels Henze, and Susanne Boll. 2009. Supporting Map-Based Wayfinding with Tactile Cues. *ACM International Conference Proceeding Series*. <https://doi.org/10.1145/1613858.1613888>
- [24] Yatharth Singhal, Haokun Wang, Hyunjae Gil, and Jin Ryong Kim. 2021. Mid-Air Thermo-Tactile Feedback using Ultrasound Haptic Display. In *Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology (Osaka, Japan) (VRST '21)*. Association for Computing Machinery, New York, NY, USA, Article 28, 11 pages. <https://doi.org/10.1145/3489849.3489889>
- [25] B. Son and J. Park. 2018. Tactile Sensitivity to Distributed Patterns in a Palm. In *Proceedings of the 20th ACM International Conference on Multimodal Interaction*. ACM, New York, NY, USA, 408–415. <https://dl.acm.org/doi/abs/10.1145/3242969.3243030> Available online: ACM Digital Library.
- [26] Ruxandra Tapu, Mocanu Bogdan, and Titus Zaharia. 2018. Wearable assistive devices for visually impaired: A state of the art survey. *Pattern Recognition Letters* 137 (10 2018). <https://doi.org/10.1016/j.patrec.2018.10.031>
- [27] Vybrionics. [n.d.]. VG0832022D Datasheet. *Vybrionics Technical Documentation* ([n.d.]). Available at: <https://www.vybrionics.com/wp-content/uploads/datasheet-files/Vybrionics-VG0832022D-datasheet.pdf>, Accessed: 15 January 2025.
- [28] X. Zhang, T. Yang, D. Zhang, and N. F. Lepora. 2024. TacPalm: A Soft Gripper with a Biomimetic Optical Tactile Palm for Stable Precise Grasping. *IEEE Sensors Journal* (2024). <https://ieeexplore.ieee.org/abstract/document/10706767/> Available online: IEEE Xplore.